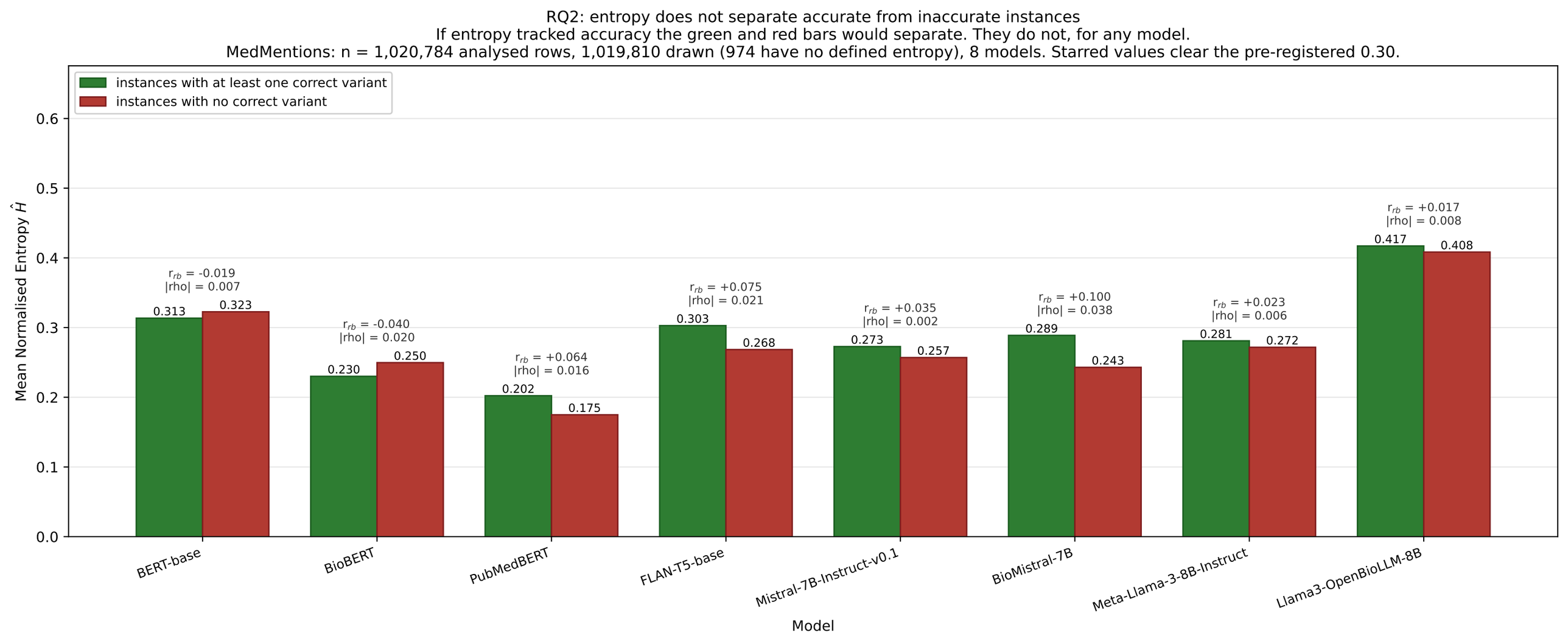
RQ2: entropy does not separate accurate from inaccurate instances
If entropy tracked accuracy the green and red bars would separate. They do not, for any model.
MedMentions: n = 1,020,784 analysed rows, 1,019,810 drawn (974 have no defined entropy), 8 models. Starred values clear the pre-registered 0.30.
instances with at least one correct variant
instances with no correct variant
Mean Normalised Entropy $\hat{H}$
0.0
0.1
0.2
0.3
0.4
0.5
0.6
r_{rb} = -0.019
|rho| = 0.007
0.313
0.323
r_{rb} = -0.040
|rho| = 0.020
0.230
0.250
r_{rb} = +0.064
|rho| = 0.016
0.202
0.175
r_{rb} = +0.075
|rho| = 0.021
0.303
0.268
r_{rb} = +0.035
|rho| = 0.002
0.273
0.257
r_{rb} = +0.100
|rho| = 0.038
0.289
0.243
r_{rb} = +0.023
|rho| = 0.006
0.281
0.272
r_{rb} = +0.017
|rho| = 0.008
0.417
0.408
BERT-base
BioBERT
PubMedBERT
FLAN-T5-base
Mistral-7B-Instruct-v0.1
BioMistral-7B
Meta-Llama-3-8B-Instruct
Llama3-OpenBioLLM-8B
Model